

春三周作业3

1.

R A B C D Y_3 Y_2 Y_1 Y_0

0 0 0 0 0 0 1 1

0 0 0 1 0 1 0 0

0 0 1 0 0 1 0 1

0 0 1 1 0 1 1 0

0 1 0 0 0 1 1 1

0 1 0 1 1 0 0 0

0 1 1 0 1 0 0 1

0 1 1 1 1 0 1 0

1 0 0 0 1 0 1 1

1 0 0 1 1 1 0 0

Y_3

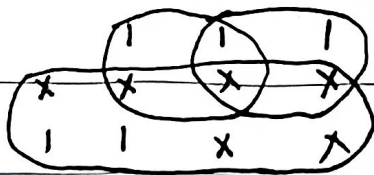
AB|CD 00 01 11 10

00

01

11

10



$$Y_3 = A + BD + BC$$

Y_2

AB\CD	00	01	11	10
00		1	1	1
01	1			
11		x	x	x
10		1	x	x

$$Y_2 = B\bar{C}\bar{D} + \bar{B}D + \bar{B}C$$

 Y_1

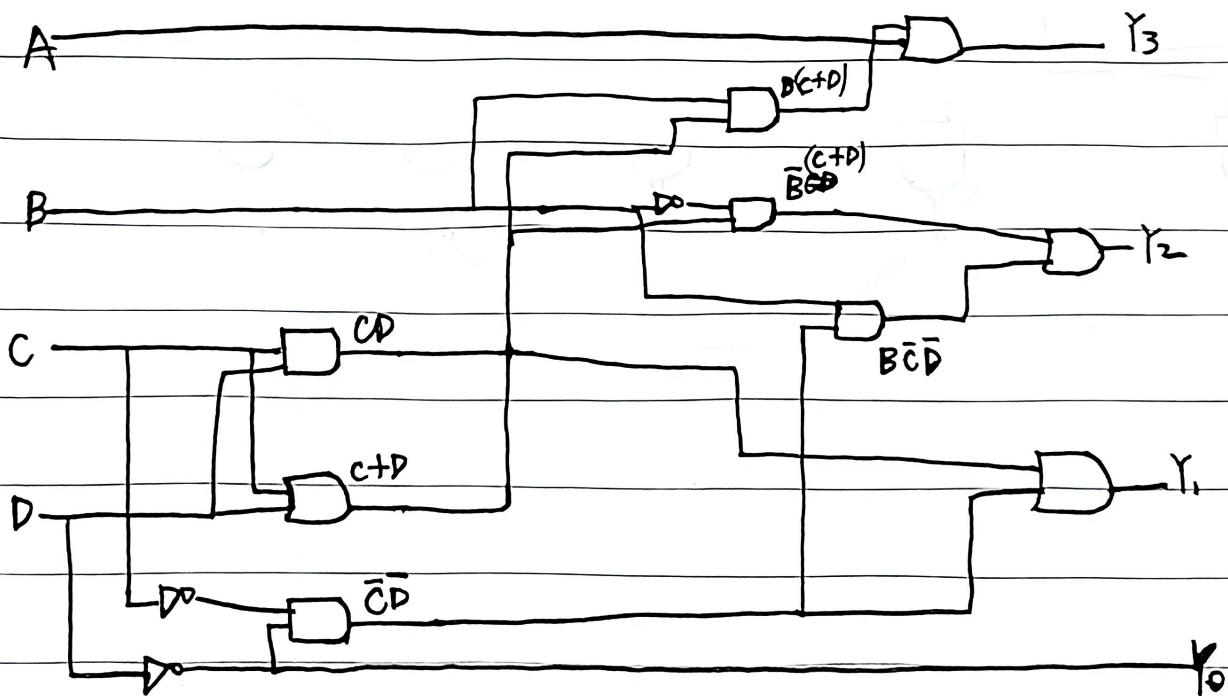
AB\CD	00	01	11	10
00		1	1	
01		1		
11		x	x	x
10		1	x	x

$$Y_1 = \bar{C}\bar{D} + CD$$

 Y_0

AB\CD	00	01	11	10
00	1			1
01	1			
11	x	x	x	x
10	1		x	x

$$Y_0 = \bar{D}$$



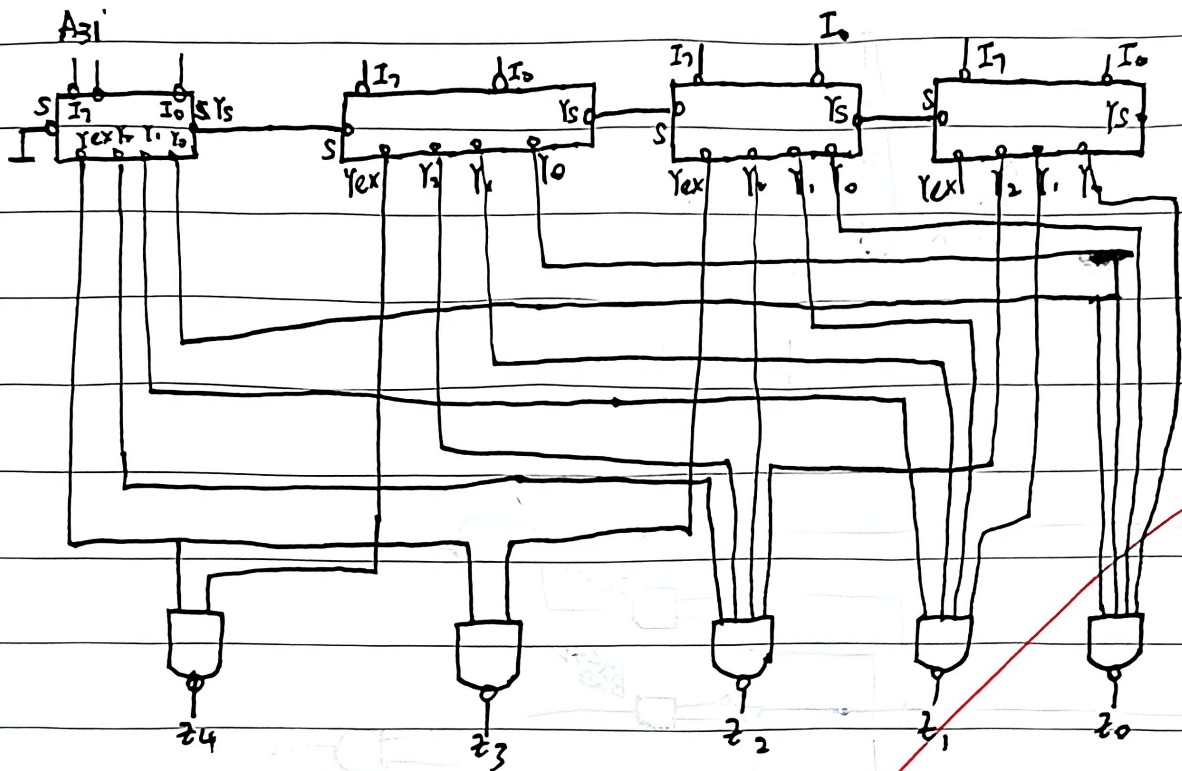
2. $\bar{V} = \bar{P}_0 \bar{D}_1 \bar{P}_2 \bar{P}_3$

$V = P_0 + P_1 + P_2 + P_3$

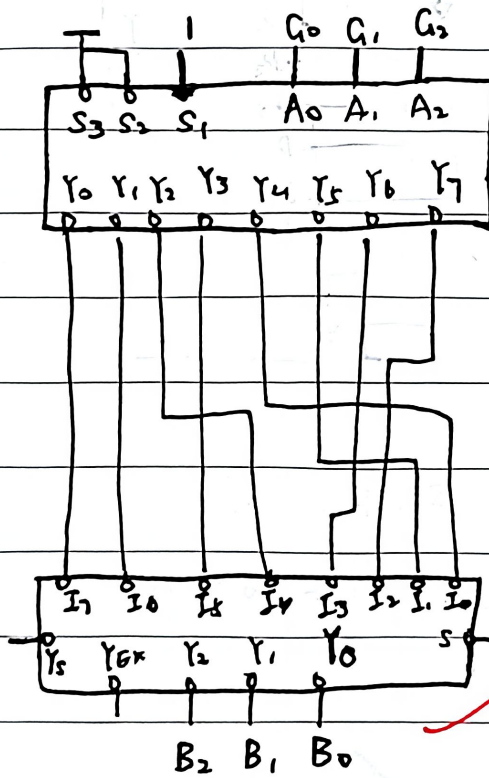
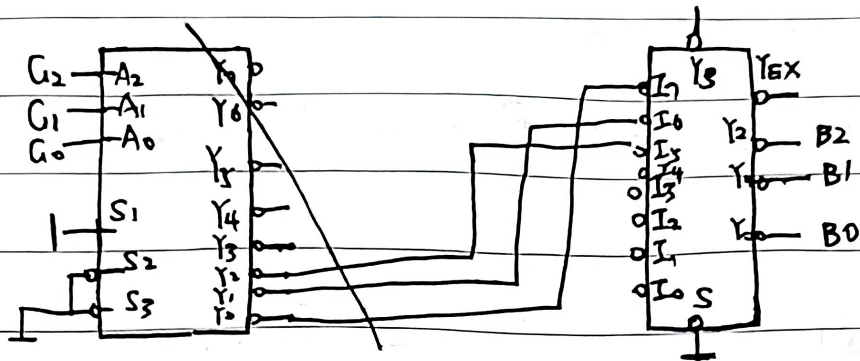
$X = P_3 + \bar{D}_3 P_2$

$Y = P_3 + \bar{D}_3 \bar{P}_2 D_1$

3.



4.

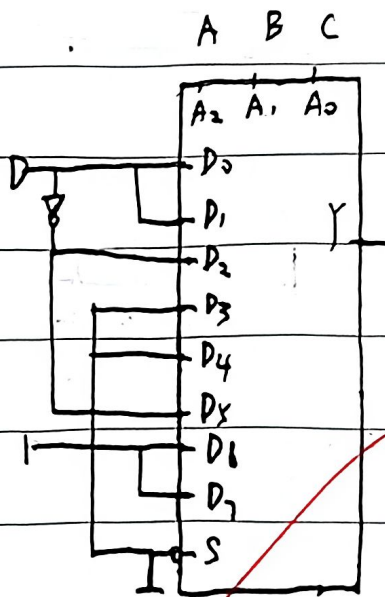
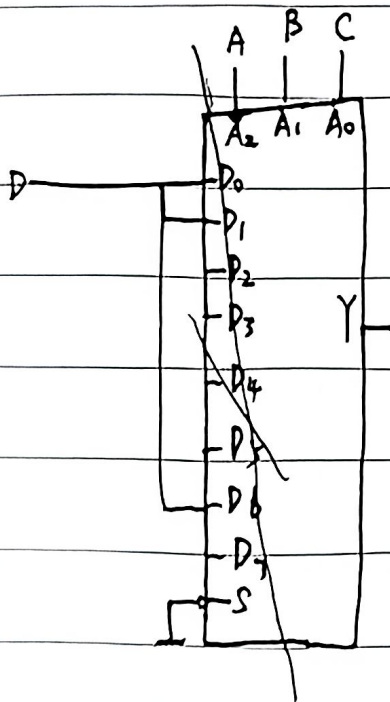


5.

$$F = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BC\bar{D} + \bar{A}B\bar{C}D + \bar{A}BCD$$

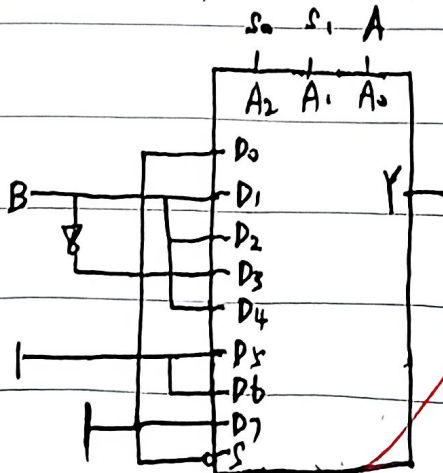
$$+ \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + \bar{A}BCD$$

$$= m_0D + m_1D + m_2\bar{D} + m_5\bar{D} + m_6\bar{D} + m_6D + m_7 \cdot 1$$



b. $Y = \bar{S}_1\bar{S}_0AB + \bar{S}_1S_0(A+B) + S_1\bar{S}_0(\bar{A}\bar{B} + \bar{A}B) + S_1S_0\bar{A}$

$$= (m_1 + m_2 + m_4)B + m_3\bar{B} + (m_5 + m_4) \cdot 1 + (m_0 + m_7) \cdot 0$$



7.

$$Y_1 = m_4 + m_6 + m_7$$

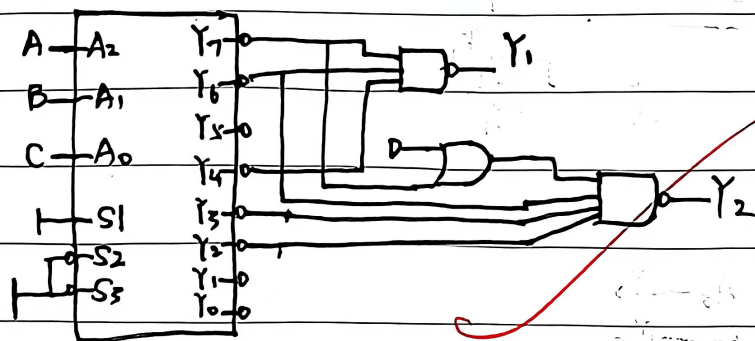
$$= \overline{m_4} \overline{m_6} \overline{m_7}$$

$$Y_2 = ABC\overline{D} + \overline{A}B + B\overline{C}$$

$$= \cancel{m_{14}} + m_4 + \cancel{m_5} + \cancel{m_6} + \cancel{m_7} + m_{12} + m_{13}$$

$$m_7 \overline{D} + m_3 + m_2 + m_6$$

$$= \overline{m_2} \overline{m_3} \overline{m_6} (\overline{m_7} + D)$$

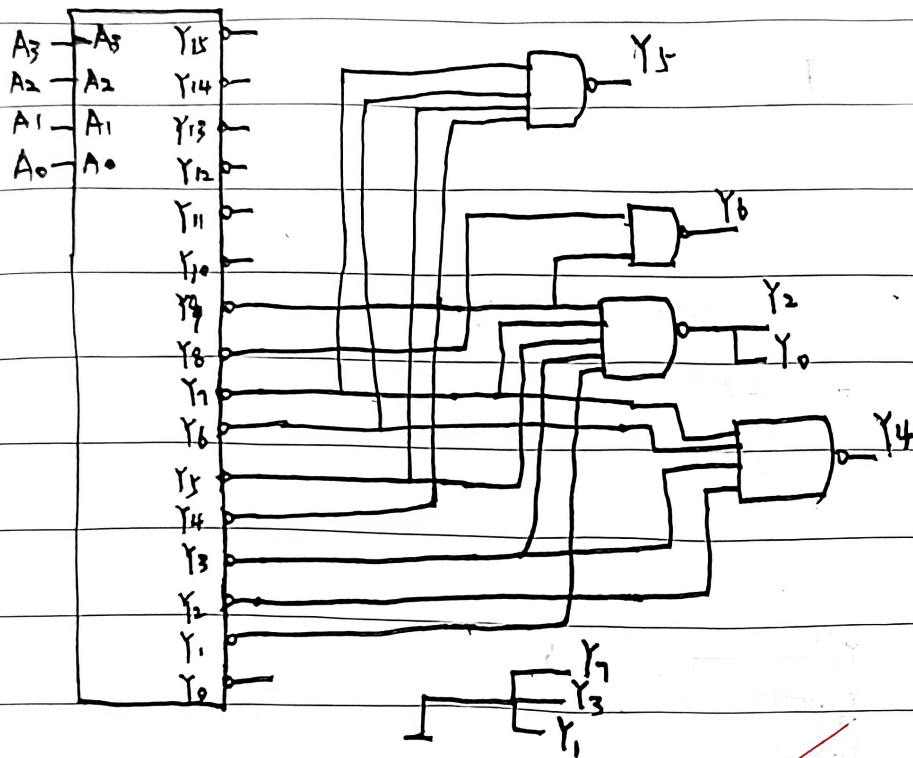


8.

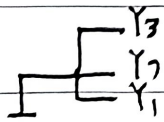
$$ABCD \text{ is } Y_7 Y_6 Y_5 Y_4 Y_3 Y_2 Y_1 Y_0$$

A ₃	A ₂	A ₁	A ₀	Y ₇	Y ₆	Y ₅	Y ₄	Y ₃	Y ₂	Y ₁	Y ₀	
0	0	0	0	0	0	0	0	0	0	0	0	Y ₇ = 0
0	0	0	1	0	0	0	0	0	1	0	1	Y ₆ = m ₈ + m ₉
0	0	1	0	0	0	0	1	0	0	0	0	Y ₅ = m ₄ + m ₅ + m ₆ + m ₇
0	0	1	1	0	0	0	1	0	1	0	1	Y ₄ = m ₂ + m ₃ + m ₆ + m ₇
0	1	0	0	0	0	1	0	0	0	0	0	Y ₃ = 0
0	1	0	1	0	0	1	0	0	1	0	1	Y ₂ = m ₁ + m ₃ + m ₅ + m ₇ + m ₉
0	1	1	0	0	0	1	1	0	0	0	0	Y ₁ = 0
0	1	1	1	0	0	1	1	0	1	0	1	Y ₀ = m ₁ + m ₃ + m ₅ + m ₇ + m ₉
1	0	0	0	0	1	0	0	0	0	0	0	
1	0	0	1	0	1	0	0	0	1	0	1	

01



2)



$A_3 - Y_6$
 $A_2 - Y_5$
 $A_1 - Y_4$
 $A_0 - Y_0$

95